

Customer Retention & RFM Analysis for Online Retail Customers

1. Project Overview

Analyze online retail customer behavior to understand retention trends and identify high-value customer segments for targeted marketing and retention strategies.

2. Dataset Description

- Rows: 1067371
- Columns: 13
- Missing Data:
 - 4382 values in *Description* column
 - 243007 values in *Customer ID*
- Key Features:
 - Order-level retail data (*Invoice, Invoice Date, Quantity*)
 - Customer details (*Country, Order Size, Customer ID, Customer Type, Sales, First Purchase Date, Quantity, Price, Description*)
 - Date: 2009 - 2012

3. Business Questions

- How long do customers stay active after their first purchase?
- Which customers generate the most revenue?
- Where should marketing focus on retention and engagement?

4. Methodology

a) Data Cleaning & Preparation:

- Converted dates, handled duplicates, calculated revenue, and classified customers as one-time, repeat, or loyal.

b) Cohort Analysis:

- Grouped customers by first purchase month (cohort_month)
- Tracked monthly retention (cohort_index)
- Created a retention heatmap to visualize customer churn over time

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c) RFM Analysis:

- Calculated **Recency** (days since last purchase), **Frequency** (number of purchases), and **Monetary** (total revenue) for each customer
- Quartile-based scoring for R, F, M → combined into RFM_score
- Segmented customers: Loyal Customers, Potential Loyalists, Others, At Risk
- Revenue per segment analyzed to prioritize high-value customers

5. Key Insights

a) Cohort Analysis

- Retention drops sharply after the first month, with the majority of customers not returning beyond 3–4 months.
- Newer cohorts show slight improvement, indicating increased engagement over time.
- Early churn represents a major opportunity for targeted retention campaigns.

b) Customer Geography & Seasonality

- **90% of sales** come from the UK, followed by Germany and France.
- Sales begin increasing in **August**, peaking in **October**, indicating clear seasonal trends.

c) Customer Type

- From customer_type, **most customers are repeat buyers (≤10 purchases)**, suggesting consistent but moderate engagement.

d) RFM Analysis

- **Loyal Customers** contribute ~11.6M in revenue, highlighting their strategic importance.
- **Potential Loyalists** and **Others** represent moderate revenue, with potential for growth through marketing initiatives.
- **At Risk** customers contribute the least revenue (~1M) and may require re-engagement strategies.

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- No “Champions” exist, indicating few high-frequency, high-value repeat buyers.

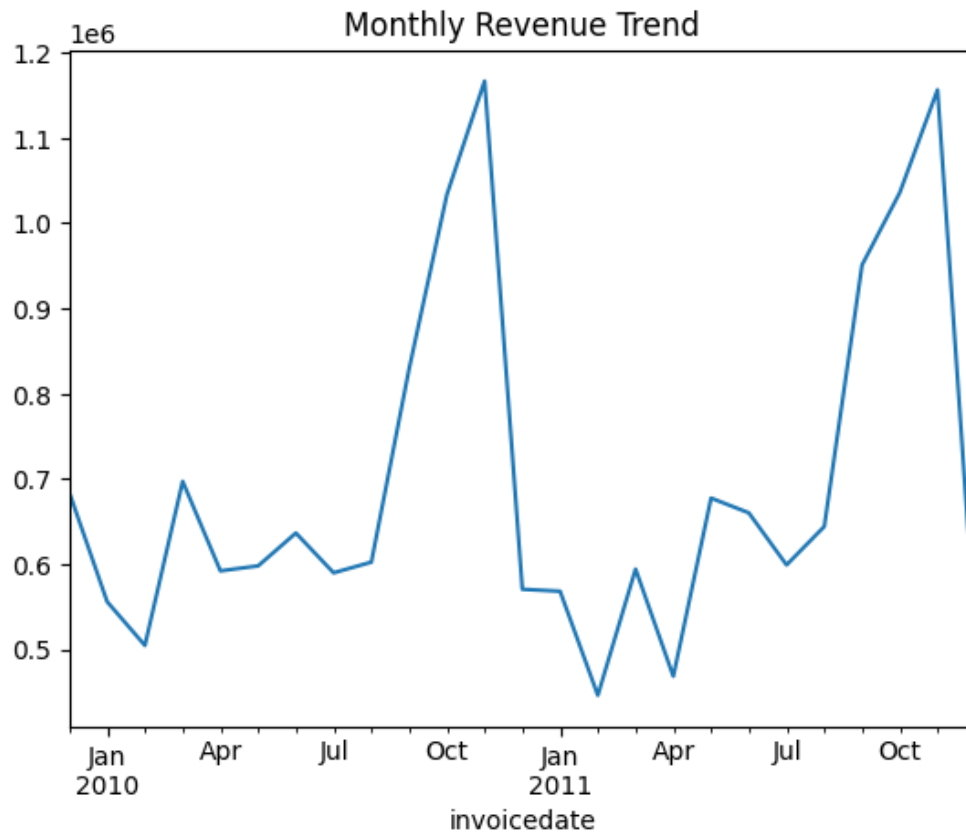
6. Recommendations

- Focus retention campaigns on Loyal Customers to maximize revenue.
- Implement targeted campaigns for Potential Loyalists to convert them into Loyal Customers.
- Monitor early churn trends and design incentives for first-time buyers to encourage repeat purchases.
- Consider seasonal promotions aligned with August–October peak months to maximize sales.
- Explore localized marketing strategies in high-revenue regions (UK, Germany, France).

7. Visualizations

Customer Retention & RFM Analysis for Online Retail Customers

→ Monthly Revenue Trend



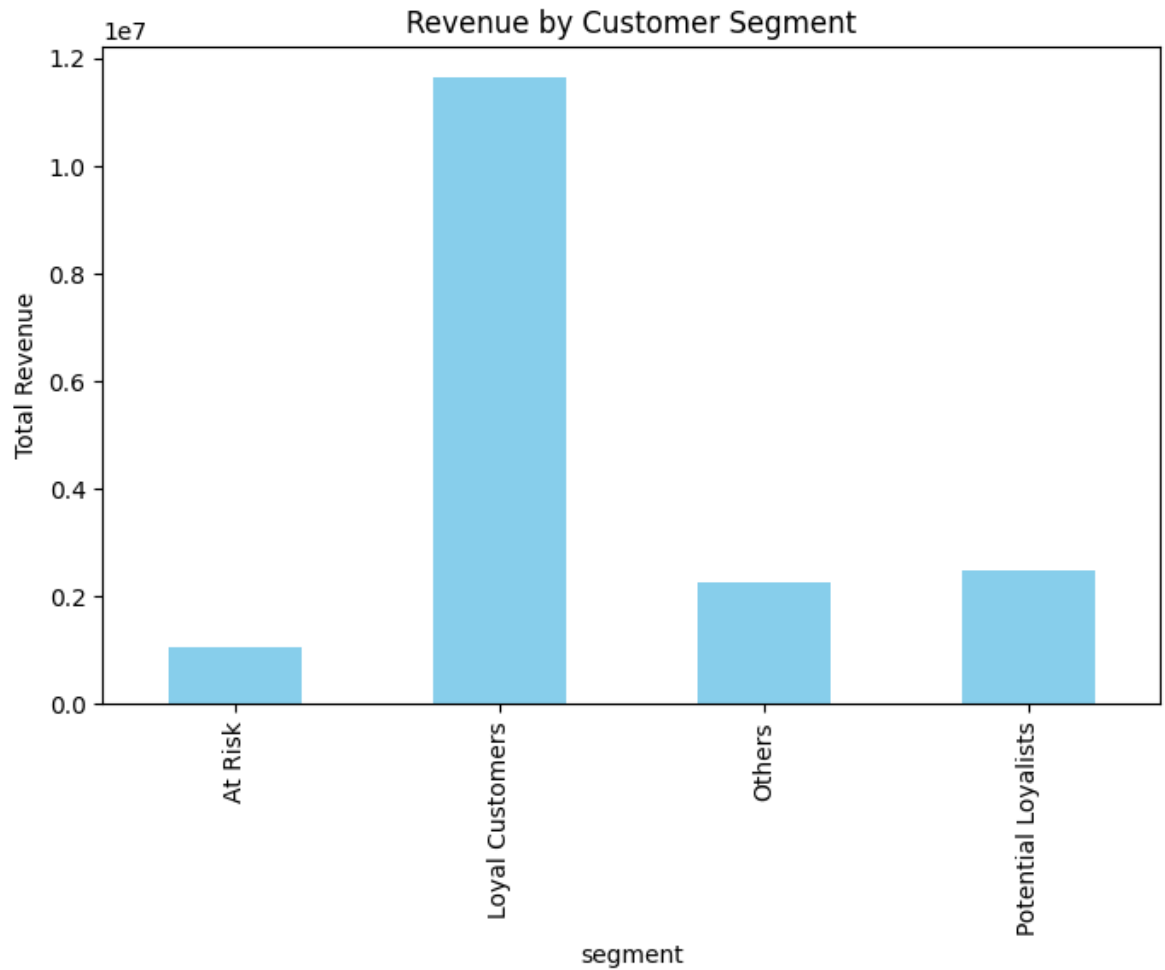
Customer Retention by Cohort

This heatmap visualizes customer retention rates across different cohorts and time periods. The vertical axis lists cohorts from 2009-12 to 2011-12. The horizontal axis shows months since first purchase from 1 to 25. A color scale on the right indicates retention percentages from 0.0 (dark blue) to 1.0 (light yellow). Numerical values are printed on each cell.

Cohort Month	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
2009-12	100%	85%	33%	43%	38%	36%	38%	34%	34%	36%	42%	50%	38%	28%	24%	30%	26%	30%	28%	26%	26%	32%	30%	41%	20%
2010-01	100%	21%	31%	31%	26%	30%	26%	23%	28%	32%	30%	17%	22%	18%	19%	15%	23%	20%	19%	20%	24%	19%	25%	6%	
2010-02	100%	24%	22%	29%	25%	20%	19%	29%	25%	28%	11%	13%	15%	17%	12%	20%	16%	16%	14%	23%	23%	16%	6%		
2010-03	100%	19%	23%	24%	23%	20%	25%	30%	28%	11%	12%	14%	20%	16%	20%	17%	17%	16%	18%	20%	21%	8%			
2010-04	100%	19%	19%	16%	18%	22%	28%	26%	11%	11%	7%	14%	14%	16%	16%	16%	14%	15%	18%	22%	21%	6%			
2010-05	100%	16%	17%	17%	18%	26%	21%	13%	6%	8%	11%	13%	15%	15%	10%	13%	14%	17%	15%	5%					
2010-06	100%	17%	19%	20%	23%	29%	13%	9%	8%	12%	11%	14%	15%	12%	11%	12%	13%	20%	5%						
2010-07	100%	16%	18%	30%	29%	14%	11%	15%	15%	11%	13%	15%	13%	13%	19%	17%	24%	8%							
2010-08	100%	20%	30%	32%	17%	12%	10%	12%	14%	13%	13%	12%	15%	19%	18%	20%	7%								
2010-09	100%	23%	23%	12%	9%	10%	14%	10%	13%	13%	12%	10%	22%	16%	16%	5%									
2010-10	100%	26%	15%	12%	9%	8%	13%	14%	11%	9%	11%	13%	19%	23%	7%										
2010-11	100%	18%	9%	10%	8%	9%	13%	10%	9%	9%	11%	14%	26%	6%											
2010-12	100%	9%	5%	9%	12%	7%	5%	9%	5%	3%	9%	20%	3%												
2011-01	100%	17%	21%	20%	21%	15%	15%	13%	11%	21%	25%	11%													
2011-02	100%	16%	15%	19%	22%	15%	15%	15%	15%	18%	6%														
2011-03	100%	18%	22%	20%	22%	15%	21%	20%	24%	9%															
2011-04	100%	25%	20%	20%	18%	24%	18%	25%	11%																
2011-05	100%	23%	24%	16%	22%	21%	26%	12%																	
2011-06	100%	23%	21%	27%	20%	29%	8%																		
2011-07	100%	22%	30%	27%	34%	16%																			
2011-08	100%	27%	31%	26%	17%																				
2011-09	100%	27%	38%	15%																					
2011-10	100%	32%	16%																						
2011-11	100%	14%																							
2011-12	100%																								

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→ Revenue by Customer Segment



Customer Retention & RFM Analysis for Online Retail Customers

→ Total Revenue by Customer Type

